TE 458 — Coverage Planning: Tutorial 1 Practice Set (Variant C)

4 more numeric variants of Tutorial 1's problems, same methods, new numbers

These are freshly parameterised variants of Tutorial 1's four problems — same methods and formulas (see the companion Formula Sheet), different input numbers — for additional self-test practice. Propagation-model coefficients are given directly in each question, exactly as they would be handed to you on an exam.

Problem 1 — 5G NR Downlink (variant)

In a 5G NR system operating at 1.8 GHz, the base station transmitter power is 46 dBm, with a feeder loss of 3 dB and an antenna gain of 15 dBi. For the UE receiver, the noise figure is 8 dB with a noise floor of –93 dBm, and the required SNR (E_b/N_0) for QPSK is 6 dB.

(Boltzmann's constant: $k = 1.38\times10^{-23}$ J/K, T_0 = 290 K. Additional margins: shadow-fading margin = 6 dB, interference margin = 3 dB. Propagation model: MAPL = 126.0 + 35.0 $\log_{10}(R)$.)

a. Calculate the EIRP.

b. Calculate the bandwidth.

c. Compute the receiver sensitivity.

d. Including the shadow-fading and interference margins above, find the Maximum Allowable Path Loss for the downlink.

e. Using the given propagation model, determine the cell radius (R).

Problem 2 — 5G NR Uplink (variant)

A 5G NR uplink operates at f = 1.8 GHz. The user equipment transmits P_{TX} = 125 mW with antenna gain G_{TX} = 0 dBi and no feeder loss. The gNB receiver parameters are:

Parameter	Value
Receive antenna gain, G_{RX}	16 dBi
Receiver-side loss, L_{RX}	1 dB
Noise figure, NF	7 dB
Bandwidth, B	10 MHz

To support 16-QAM modulation, the required SNR is 9 dB. Additional losses and margins: body loss = 2 dB, building penetration loss = 6 dB, fade + interference margin = 9 dB. Assume T_0 = 295 K.

a. Calculate the EIRP.

b. Determine the free-space path loss at a reference distance of 100 m.

c. Assuming T_0 = 295 K, determine: i. thermal noise spectral density; ii. thermal noise power in 10 MHz; iii. receiver noise floor; iv. receiver sensitivity.

d. Compute the maximum allowable path loss (MAPL).

e. Using the log-distance model with n = 3.0 and reference distance d_0 = 100 m, determine: i. the coverage radius; ii. the coverage area assuming a hexagonal cell.

f. State and quantify the effect on coverage area if receiver sensitivity improves by 3 dB.

Problem 3 — UMTS System Design (variant, handheld device)

As part of a UMTS system design, the following parameters are given to perform a link budget analysis for a 32-kbps handheld voice/data service (unlike the laptop data-card case in earlier problem sets, this device is held against the body, so body loss is nonzero here):

Parameter	Value
Mobile TX power (dBm)	23
Antenna gain (dBi)	1
Body loss (dB)	2
Thermal noise density (dBm/Hz)	–174
Receiver noise figure (dB)	5.5
Interference margin (dB)	3.5
Required E_b/N_0 (dB)	2.5
Base station antenna gain (dBi)	16
Base station feeder and connector losses (dB)	2
Fast fading margin (dB)	4
Log-normal fade margin (dB)	7
Building penetration loss (dB)	13
Soft handover gain (dB)	2
Chip rate, M_{cps}	3.84×10^6

a. Perform link budget calculations for the 32 kbps handheld service in the uplink direction, considering 15 users.

b. What is the maximum allowable path loss? (Propagation model: MAPL = 134.0 + 34.5 $\log_{10}(R)$.)

c. What is the coverage radius of the base station?

Problem 4 — WCDMA Macro-Cell Analysis (variant)

You are planning the uplink link budget for a WCDMA macro-cell in a dense urban area supporting a 256 kbps video streaming service. Given:

Parameter	Value
Mobile transmit power	22 dBm
Mobile antenna gain	1.5 dBi
Body loss	3.5 dB
Base station receiver noise figure	5.2 dB
Chip rate	3.84 Mcps
Video service bit rate	256 kbps
Required E_b/N_0	5.2 dB
Thermal noise density	–174 dBm/Hz
Interference margin (75% cell loading)	8 dB
Slow fading margin (97% edge coverage probability)	9.5 dB
Soft handover gain	+1.8 dB

a. Compute the following step-by-step, showing all intermediate formulas: i. EIRP; ii. receiver noise power; iii. total noise-plus-interference power; iv. processing gain; v. effective receiver sensitivity; vi. Maximum Allowable Path Loss; vii. cell radius R using the propagation model $PL_{max} = 136.5 + 35.8 \log_{10}(R)$.

b. Perform a quantitative impact analysis on the hexagonal cell coverage area $A = (3\sqrt{3}/2)R^2$:

i. If the service bit rate is increased to 512 kbps (same E_b/N_0), calculate the new radius and percentage reduction in coverage area.

ii. If the interference margin rises to 11 dB due to higher network load, determine the new MAPL, radius, and area. Discuss the trade-off with system capacity.